

FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Main)-2023

TEST DATE: 12-11-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

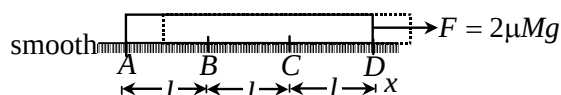
Sol. If the block has moved by distance x ,

$$Ma = 2\mu Mg - \left[\frac{\mu Mg}{3} + \frac{2\mu Mg}{3} + \frac{3\mu Mg}{3l}(l-x) \right]$$

$$a = 2\mu g - \left(\frac{\mu g}{3} + \frac{2\mu g}{3} + \frac{3\mu g}{3l}(l-x) \right)$$

$$\frac{da_1}{dx} = \frac{3\mu g}{3l} \quad [\text{for } 2^{\text{nd}} \text{ and } 3^{\text{rd}} \text{ cells similarly}]$$

$$\tan \theta_1 : \tan \theta_2 : \tan \theta_3 = 3 : 2 : 1$$



2. A

Sol. $\vec{F} = I \int d\vec{\ell} \times \vec{B}$, $B \approx \frac{\mu_0 I}{d} \Rightarrow F \propto I^2 \mu_0$

$$\Rightarrow [MLT^{-2}] = \left(\frac{Q}{T} \right)^2 \mu_0$$

$$\Rightarrow \mu_0 = [MLQ^{-2}]$$

3. A

Sol. $L = m[\vec{r} \times (\vec{\omega} \times \vec{r})] + m[\vec{r} \times (\vec{\omega} \times \vec{r})] = 2m[\vec{r} \times \ell \omega \sin \theta] = 2m\omega \ell^2 \sin \theta$

$$\text{Torque} = \frac{dL}{dt} = \vec{\omega} \times \vec{J} = 2m\omega \ell^2 \sin(90 - \theta) \sin \theta = 2m\omega \ell^2 \cos \theta \sin \theta$$

4. B

$$\text{Sol. } \frac{dm}{dt} = kt^2 \Rightarrow m = \frac{kt^3}{3} + 2$$

$$m \frac{dv}{dt} + v \frac{dm}{dt} = mg \Rightarrow m \frac{dv}{dt} + vkt^2 = mg$$

$$\frac{dv}{dt} + \frac{kt^2}{m}v = g \Rightarrow \frac{dv}{dt} + \frac{3kt^2}{kt^3 + 6}v = g \text{ which is linear differential equation}$$

$$v(kt^3 + 6) = \int g(kt^3 + 6)dt + c, t = 0, v = 0 \Rightarrow c = 0$$

$$v = \frac{g(kt^4 + 24t)}{4(kt^3 + 6)}$$

Put 2 gm , $k = 12 \text{ gm/s}^3$ and $g = 100 \text{ cm/s}^2$ and $t = 1$

$$v = \frac{1000(12 + 24)}{4(12 + 6)} = 500 \text{ cm/sec}$$

5. C

$$\text{Sol. } \vec{F}(x) = \left(\frac{2}{x^2} - \frac{x^2}{2} \right) \hat{e}_x, x \geq 0$$

Work done in a cycle is zero so conservative

$\vec{r} \times \vec{F} = 0$, so angular momentum is conserved

$$\text{For equilibrium point } F = 0 \Rightarrow F = \left(\frac{2}{x^2} - \frac{x^2}{2} \right) = 0 \Rightarrow x = -\sqrt{2}, \sqrt{2} \quad x > 0 \text{ so equilibrium point is } \sqrt{2}$$

$$\text{To check stable and unstable equilibrium point } \frac{dF}{dx} = 0 \Rightarrow \frac{-4}{x^3} - \frac{2x}{2} = 0$$

$$\text{At } x = \sqrt{2}, \frac{dF}{dx} = 0 \Rightarrow \frac{-4}{x^3} - \frac{2x}{2} = -\frac{4}{2\sqrt{2}} - \sqrt{2} = -ve$$

So, it is unstable point so particle moves away from point $x = \sqrt{2}$

So, it is obvious that it will moves towards origin.

6. A

Sol. Speed is never less than v_0 in x whereas in y it is less than v_0 first and then increases $t v_0$ so, y take more time.

7. D

Sol. Friction will produce heat energy and so the total energy will not be PE + KE (or all of the PE will not turn into KE). At a constant velocity there is no gain of KE. On a horizontal surface there is no change of PE. Falling freely, the PE turns solely into KE.

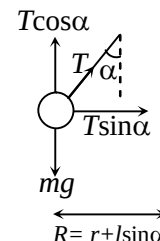
8. C

$$\text{Sol. } T \sin \alpha = m\omega^2 R$$

$$T \cos \alpha = mg$$

$$\tan \alpha = \frac{\omega^2 R}{g}$$

$$\omega = \sqrt{\frac{g \tan \alpha}{R}} = \sqrt{\frac{g \tan \alpha}{r + \ell \sin \alpha}}$$

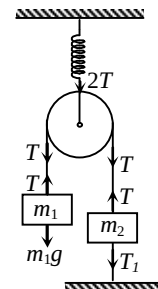


9. A

Sol. $T = m_1 g$
 $kx = 2T = 2m_1 g$

$$x = \frac{2m_1 g}{k}$$

$$\text{Energy stored} = \frac{1}{2} kx^2 = \frac{1}{2} k \frac{4m_1^2 g^2}{k^2} = \frac{2m_1^2 g^2}{k}$$



10. B

Sol. The rod will rotate about point A with angular acceleration:

$$\alpha = \frac{\tau}{I} = \frac{Fx}{\frac{m\ell^2}{3}} = \frac{3Fx}{m\ell^2}$$

$$\therefore a = \frac{\ell}{2} \alpha = \frac{3}{2} \frac{Fx}{m\ell} \text{ or } a \propto x$$

i.e., a-x graph is a straight line passing through origin.

11. A

Sol. $v \cos 60^\circ = \frac{3v_0}{2} \cos 30^\circ, v = \frac{3\sqrt{3}}{2} v_0$

12. C

Sol. $v_1 = \frac{v_0 t}{40}, S_1 = \frac{v_0}{40} \frac{t^2}{2} \Big|_{30}^t$

$$v_2 = \frac{v_0 t}{10} - 3v_0, S_2 = \frac{v_0 t^2}{20} - 3v_0 t \Big|_{30}^t$$

$$S_1 = S_2$$

$$\Rightarrow t = 60 \text{ sec}$$

13. B

Sol. $v_{\min} = \sqrt{g(y + \sqrt{x^2 + y^2})} = 30 \text{ m/s}$

$$\tan \theta = \frac{v^2}{gx} = 3$$

14. C

Sol. $N_1 + f_2 = 2mg$

... (i)

$$N_2 = f_1$$

... (ii)

$$f_1 = \mu N_1$$

... (iii)

$$f_2 = \mu N_2$$

... (iv)

Taking moment about corner B of plate

$$(mg + N_2)R = N_1 R$$

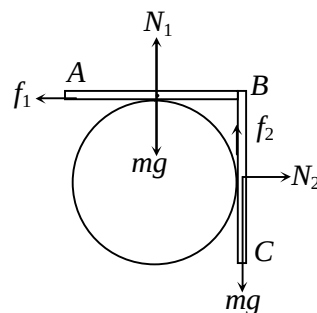
$$mg + N_2 = N_1$$

... (v)

From (i), (ii) & (v),

$$f_1 + f_2 = mg$$

... (vi)



From (iii), (iv) & (vi),

$$\mu^2 + 2\mu - 1 = 0 \text{ or}$$

$$\mu = \sqrt{2} - 1$$

15. A

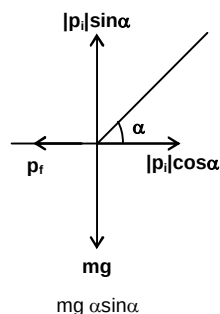
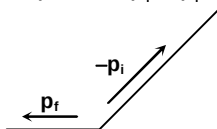
Sol. Centre of mass of arc is $\frac{2R \sin \alpha}{3\alpha}$

$$m_1 = 2m, m_2 = m$$

$$\text{centre of mass} = \frac{\frac{2m \left(2R \sin \left(\frac{\pi}{3} \right) \right)}{\left(3 \frac{\pi}{3} \right)} + m \left(\frac{-2R \sin \left(\frac{\pi}{6} \right)}{2 \left(\frac{\pi}{6} \right)} \right)}{3m} = \frac{2R(\sqrt{3} - 1)}{3}$$

16. A

Sol. impulse = $\vec{p}_f - \vec{p}_i$



17. A

Sol. $t = 0$ to $t = 1$ sec

$$v = 4t \text{ m/s}$$

$$s = 2t^2 \text{ m}$$

$$s(t = 0) = 0 \text{ m}, s(t = 1) = 2 \text{ m}$$

$$v_{\text{average}} = \frac{\Delta s}{\Delta t} = 2 \text{ m/s}$$

$t = 1$ to $t = 2$ sec

$$v = 2 \text{ m/s}$$

$$s = 2t \text{ m}$$

$$s(t = 1) = 2 \text{ m}, s(t = 2) = 4 \text{ m}$$

$$v_{\text{average}} = 2 \text{ m/s}$$

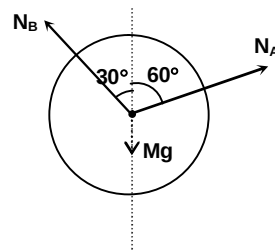
18. C

Sol. A cylinder of mass M

For equilibrium $N_A \cos 60^\circ + N_B \cos 30^\circ = Mg$

$$\text{and } N_A \sin 60^\circ = N_B \sin 30^\circ$$

$$\text{On solving } N_B = \sqrt{3}N_A; N_A = \frac{Mg}{2}$$



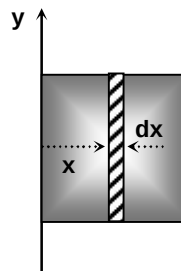
19. C

Sol. $dM = \sigma_0 \left(1 - \frac{x}{a}\right) dx$

$$M = \frac{\sigma_0 a}{2}$$

$$\Rightarrow d(MOI) = dm x^2$$

$$MOI = \int dM x^2 = \frac{Ma^2}{6}$$



20. C

Sol. Minimum stopping distance = s

Force of friction = μmg

Work done against the friction $W = \mu mgs$

Initial kinetic energy of the toy cart = $\left(\frac{p^2}{2m}\right)$

$$\therefore \mu mgs = \left(\frac{p^2}{2m}\right)$$

$$\frac{s_1}{s_2} = \left(\frac{m_2}{m_1}\right)^2$$

SECTION – B

21. 3

Sol. Let the shell hit the plane at $p(x, y)$, the range being $AP = R$, as shown in the figure. The equation for the projectile's motion is

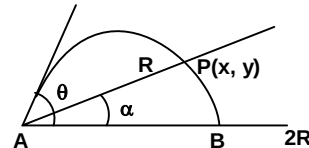
$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} \quad \dots(i)$$

$$\text{Now } y = R \sin \alpha \quad \dots(ii)$$

$$x = R \cos \alpha \quad \dots(iii)$$

Using equation (ii) and equation (iii) in equation (i) and simplifying

$$R = \frac{2u^2 \cos \theta \sin(\theta - \alpha)}{g \cos^2 \alpha}$$



The maximum range is obtained by setting $\frac{dR}{d\theta} = 0$, holding u , α and g constant. This gives

$$\cos(2\theta - \alpha) = 0 \text{ or } 2\theta - \alpha = \frac{\pi}{2}$$

22. 5

Sol. for block A

$$T = \mu m_A g + m_A \omega^2 R$$

For block B

$$T + \mu(m_A + m_B)g + \mu m_A g = m_B \omega^2 R$$

$$\omega = \sqrt{\frac{\mu g(3m_A + m_B)}{R(m_B - m_A)}}$$

23. 120

 Sol. As $\Delta p_x = 0$, $\Delta p_y = 0$

For x direction,

$$\Rightarrow 2m \times \frac{v \cos \theta}{2} + mv \frac{\cos \theta}{2} = 3m \times \frac{v}{3}$$

$$\Rightarrow 2mv \frac{\cos \theta}{2} = mv \Rightarrow \frac{\cos \theta}{2} = \frac{1}{3} = \cos 60^\circ$$

$$\frac{\theta}{2} = 60^\circ \Rightarrow \theta = 120^\circ$$

24. 125

 Sol. Kinetic energy $E = \frac{1}{2}mv^2$

$$\therefore \frac{\Delta E}{E} \times 100 = \frac{v'^2 - v^2}{v^2} \times 100 = [(1.5)^2 - 1] \times 100$$

$$\frac{\Delta E}{E} \times 100 = 125\%$$

25. 16

 Sol. $v = \sqrt{v_x^2 + v_y^2}$

$$v_x = 10 - 10 \cos 5t, \quad v_y = +10 \sin 5t$$

$$v = 20 \sin\left(\frac{5t}{2}\right)$$

$$S = \int_0^t 20 \sin\left(\frac{5t}{2}\right) dt, \quad \omega t = 2\pi$$

$$\text{So, } S = 16 \text{ m}$$

26. 30

 Sol. A begins to climb B if $F \cos 60^\circ \geq Mg \sin 60^\circ$

$$F \geq \sqrt{3} \text{ Mg}$$

$$F \geq 30 \text{ N}$$

27. 3

$$\begin{aligned} \text{Sol. Power} &= \frac{\text{K.E.}}{\text{time}} = \frac{n \frac{1}{2}mv^2}{t} = \frac{3600}{60} \left(\frac{1}{2}\right) (0.2) \left(\frac{500}{3}\right)^2 \\ &= \left(\frac{500}{3}\right) \times 10^3 \text{ watt} \end{aligned}$$

28. 3

 Sol. $a = g(\sin \theta - \mu \cos \theta)$

$$\text{Also, } a = R\alpha, \quad \frac{Rz}{I} = \frac{R(\mu mg \cos \theta R)}{\frac{1}{2}mR^2} = 2\mu g \cos \theta$$

$$\therefore g(\sin \theta - \mu \cos \theta) = 2\mu g \cos \theta$$

$$\mu = \frac{\tan \theta}{3}$$

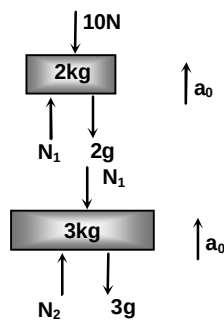
29. 7

Sol. $2a_0 = N_1 - 10 - 2g$

$$N_1 = 34 \text{ N}$$

$$3a_0 = N_2 - N_1 - 3g$$

$$N_2 = 70 \text{ N}$$



30. 8

Sol. By the conservation of energy

$$\frac{1}{2}mv_0^2 = mgh + \frac{1}{2}mv^2 \quad \dots(i)$$

By conservation of angular momentum

$$\frac{mv_0R}{2} = mvR$$

From equation (i) and equation (ii)

$$v_0 = \sqrt{\frac{8gh}{3}}$$

$$\Rightarrow k = 8$$

Chemistry

PART – B

SECTION – A

31. B

 Sol. $\begin{matrix} \text{Be} & \text{B} & \text{C} & \text{N} \\ 2s^2 & 2s^2p^1 & 2s^2p^2 & 2s^2p^3 \end{matrix}$

Be and N have stable configurations, so do not accept electron easily, hence for these elements the process is Endothermic.

So, correct order is $\text{Be} < \text{N} < \text{B} < \text{C}$

32. D

Sol. On moving down the group the IE_1 of Gr(1) elements decreases continuously hence reactivity increases.

33. D

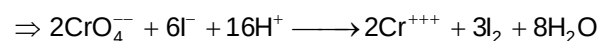
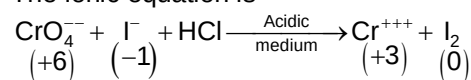
Sol. $\Delta v \times \Delta x = \frac{h}{4\pi m}$ (Using uncertainty principle)

$$\Rightarrow \Delta v = \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 1.1 \times 10^{-28} \times 3 \times 10^{-12}}$$

$$\Rightarrow \Delta v = 1.597 \times 10^5 \sim 1.6 \times 10^5 \text{ ms}^{-1}$$

34. A

Sol. The ionic equation is



or



35. A

Sol. $2\text{HI}(\text{g}) \longrightarrow \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ (follows zero order kinetics)

$$-\frac{1}{2} \frac{d[\text{R}]}{dt} = k[\text{R}]^0$$

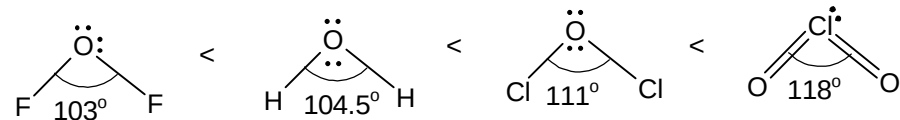
$$\Rightarrow t = \frac{[\text{R}_0] - [\text{R}]}{2k}$$

$$\Rightarrow t = \frac{1.50 - 0.30}{2 \times 0.06}$$

$$\Rightarrow t = \frac{1.20}{0.12} = 10 \text{ s}$$

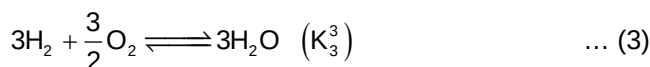
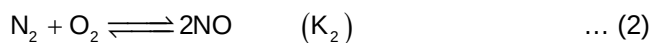
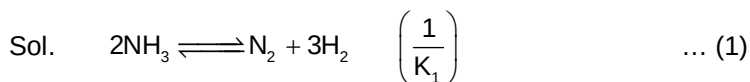
36. A

Sol.

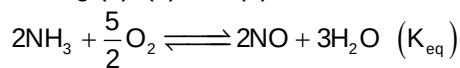


H₂O has larger bp – bp repulsion than OF₂ and OCl₂ has larger steric repulsion between two large Cl atoms while in ClO₂ the bond angle is 118° (close to 120°).

37. C

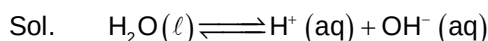


Adding (1), (2) and (3)



$$\Rightarrow K_{\text{eq}} = \frac{K_2 K_3^3}{K_1}$$

38. B



At 25°C, $[\text{H}^+] = [\text{OH}^-] = 10^{-7} \text{ M}$

$\Rightarrow K_{w_1} = 10^{-14}$

At 35°C, $[\text{H}^+] = [\text{OH}^-] = 10^{-6} \text{ M}$

$\Rightarrow K_{w_2} = 10^{-12}$

Now, $2.303 \log \frac{K_{w_2}}{K_{w_1}} = \frac{\Delta H}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$

$\Rightarrow 2.303 \log \frac{10^{-12}}{10^{-14}} = \frac{\Delta H}{2} \left(\frac{308 - 298}{308 \times 298} \right)$

$\Rightarrow 2.303 \times 2 = \frac{\Delta H}{2} \times \frac{10}{308 \times 298}$

$\Rightarrow \Delta H = 84.551 \text{ Kcal mol}^{-1}$

Since, 84.551 Kcal mol⁻¹ is the enthalpy change for the reaction

39. D

Sol. $\text{KE} = -\frac{1}{2}(\text{PE}) = -\frac{1}{2} \left(-\frac{1}{2} mkr^4 \right)$

$\text{KE} = \frac{1}{4} mkr^4 = \frac{1}{2} mv^2$

$\Rightarrow v^2 = \frac{mkr^4 \cdot 2}{4m} = \frac{1}{2} kr^4 \quad \dots (1)$

Since, $mvr = \frac{nh}{2\pi} \Rightarrow v^2 = \frac{n^2 h^2}{4\pi^2 m^2 r^2} \quad \dots (2)$

From (1) and (2)

$\frac{1}{2} kr^4 = \frac{n^2 h^2}{4\pi^2 m^2 r^2}$

$\Rightarrow r^6 \propto n^2 \Rightarrow r \propto \sqrt[3]{n}$

40. D

$$\text{Sol. } K_p = \frac{P_{\text{H}_2\text{O(g)}}^2}{P_{\text{H}_2\text{(g)}}^2} \quad \dots (1)$$

$$P_{\text{H}_2} = \frac{40}{40 + 60} \times P_T = 0.4P_T$$

$$P_{\text{H}_2\text{O}} = \frac{60}{40 + 60} \times P_T = 0.6P_T$$

$$\Rightarrow \frac{P_{\text{H}_2\text{O}}}{P_{\text{H}_2}} = \frac{0.6P_T}{0.4P_T} = \frac{3}{2} = 1.5$$

$$\therefore \text{From Eq. (1)} K_p = (1.5)^2 = 2.25$$

41. C

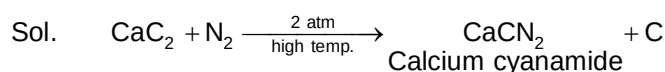
 Sol. (A) $\Rightarrow \Delta n_g > 0$, $\uparrow$ in P shifts the equilibrium $\leftarrow$

 (B) $\downarrow$ in volume in (III)rd equilibrium shifts the equilibrium $\rightarrow$ ($\Delta n_g < 0$)

(C) At constant volume no effect of addition of He

 (D) At constant pressure, addition of He shifts the equilibrium $\leftarrow$ ($\Delta n_g < 0$)

42. B



43. B

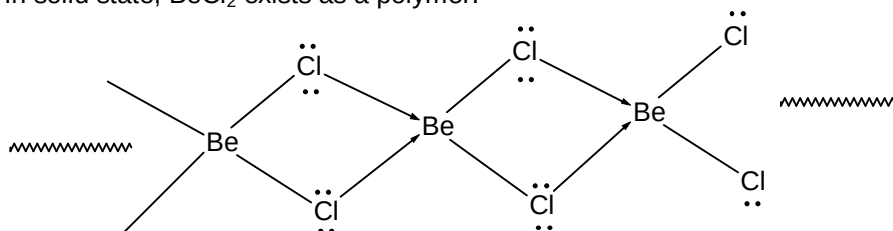
Sol. (A) On moving down the group, the size of metal ion increases, so stability of peroxides/superoxides increases.

i.e. Large cation – Large anion interactions increases.

(B) NaOH is hygroscopic in nature.

 (C), (D) Fact based on $\Delta_{\text{lattice}} \text{H}^\ominus$ and $\Delta_{\text{hyd}} \text{H}^\ominus$.

44. A

 Sol. In solid state, BeCl_2 exists as a polymer.


(four electrons are shared between three atoms, so 3c-4e type of bonds are present)

45. D

Sol. I. – 1 corner shared

II. – 2 corners shared

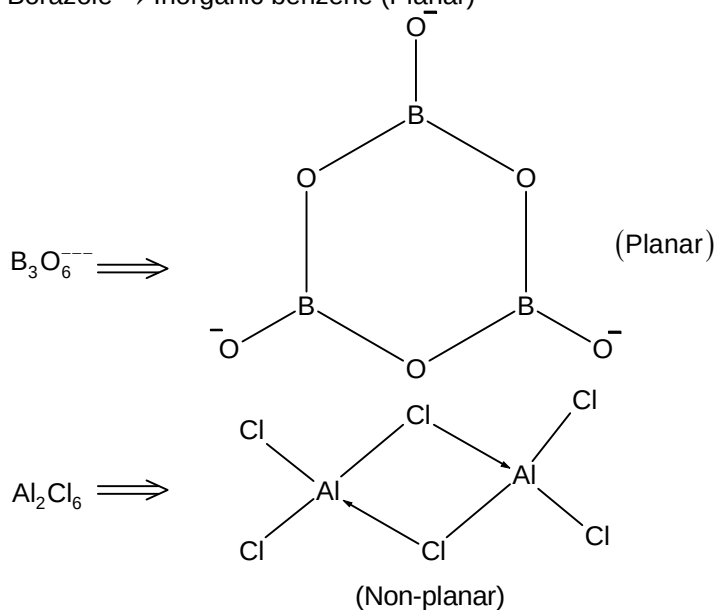
III. – One unit shares 2 corners while one unit shares 3-corners.

IV. – 2 corners shared

V. – 4 corners shared

VI – 3 corners shared

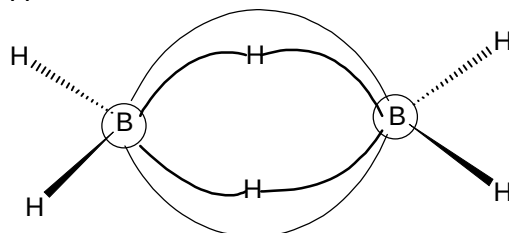
46. D
Sol. Borazon $\rightarrow$ Diamond like structure (Non-planar)
Borazole $\rightarrow$ Inorganic benzene (Planar)



47. B
Sol. $NO_2 \rightarrow$ odd electron species
 $\rightarrow$ Paramagnetic
 NO_2^+ is linear while NO_2^- and NO_2 are bent.
 $NO_2^- \rightarrow$ lp – bp repulsion
 $NO_2 \rightarrow$ bond angle is around 134°
Bond order of NO_3^- is $\frac{4}{3} = 1.33$
Bond angles, $NO_2^- (115^\circ)$, $NO_2^+ (180^\circ)$, $NO_2 (134^\circ)$

48. D
Sol. $LiH \rightarrow$ Stable
 $NaH \rightarrow$ Unstable
 $Li_2CO_3 \rightarrow$ Unstable
 $Na_2CO_3 \rightarrow$ Stable
(Around $400 - 500^\circ C$)

49. C
Sol. I. $\rightarrow$ Bond length order is $H_2^- > H_2^+ > H_2$
II. $\rightarrow$ Isoelectronic species, so identical bond order.
III. $\rightarrow$ Fact.
IV $\rightarrow NO_3^-$ (Resonance), BO_3^- (no resonance).

50. A
 Sol.


Maximum 4 – H atoms lie in a plane.

Maximum 6 – atoms (4H + 2B) lie in a plane.

Bridge (B – H) is stronger than terminal (B – H)

Terminal (H – B – H) bond angle > Bridge (H – B – H) bond angle.

SECTION – B

51. 12

 Sol. $\text{Ag}_2\text{CO}_3 (\text{s}) \rightleftharpoons 2\text{Ag}^+ (\text{aq}) + \text{CO}_3^{--} (\text{aq})$

$$K_{\text{SP}} = [\text{Ag}^+]^2 [\text{CO}_3^{--}]$$

$$\Rightarrow [\text{Ag}^+]^2 = \left[\frac{8.2 \times 10^{-12}}{7.5 / 5} \right]$$

$$\Rightarrow [\text{Ag}^+]^2 = \left[\frac{82}{15} \times 10^{-12} \right]$$

$$\Rightarrow [\text{Ag}^+] = 2.34 \times 10^{-6} \text{ M}$$

$$\text{Now } K_{\text{SP}} (\text{AgCl}) = [\text{Ag}^+] [\text{Cl}^-]$$

$$K_{\text{SP}} = 2.34 \times 10^{-6} \times \left(\frac{0.0026}{35.5} \right)$$

$$K_{\text{SP}} = 1.71 \times 10^{-10}$$

$$\Rightarrow x = 1.71, \quad y = 10$$

$$\Rightarrow x + y = 1.71 + 10 = 11.71 \sim 12$$

52. 20

 Sol. $\text{KE} = \frac{1}{2}mv^2$

$$\Rightarrow d(\text{KE}) = d\left(\frac{1}{2}mv^2\right)$$

$$\Rightarrow d(\text{KE}) = mv(dv) \quad \dots (1)$$

$$\text{Since, } dv \times dx = \frac{h}{4\pi m}$$

$$\Rightarrow dv = \frac{h}{4\pi m(dx)} \quad \dots (2)$$

From (1) and (2)

$$d(\text{KE}) = mv \cdot \frac{h}{4\pi m(dx)}$$

$$\Rightarrow d(\text{KE}) = \frac{1}{60} \times \frac{3 \times 10^8 \times 6.62 \times 10^{-34}}{4 \times 3.14 \times \frac{6.62}{3.14} \times 10^{-12}}$$

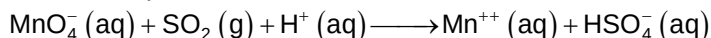
$$\Rightarrow d(\text{KE}) = \frac{1}{80} \times 10^{-14} = 1.25 \times 10^{-16} \text{ J}$$

$$\Rightarrow x = 1.25, y = 16$$

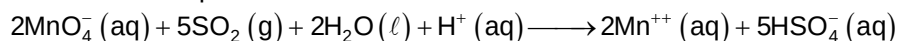
$$\Rightarrow xy = 1.25 \times 16 = 20$$

53. 7

Sol. The ionic equation involved is



The balanced equation is



$\therefore$ Sum total of coefficient(s) = 2 + 5 = 7.

54. 310

Sol. $k = Ae^{-E_a/RT}$

$$\log K = \log A - \frac{E_a}{2.303RT}$$

$$\Rightarrow \log \left(\frac{0.693}{693} \right) = \log (4 \times 10^{13}) - \frac{98.6 \times 1000}{2.303 \times 8.314 \times T}$$

or

$$\Rightarrow \log \frac{4 \times 10^{13}}{10^{-3}} = \frac{98.6 \times 1000}{2.303 \times 8.314 \times T}$$

$$\Rightarrow T = 310.1 \text{ K} \sim 310 \text{ K}$$

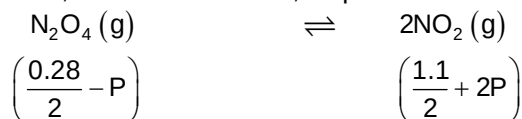
55. 64

Sol. $\text{N}_2\text{O}_4 (\text{g}) \rightleftharpoons 2\text{NO}_2 (\text{g})$

$$\begin{array}{ccc} 0.28 & & 1.1 \end{array}$$

$$K_p = \frac{(P_{\text{NO}_2})^2}{(P_{\text{N}_2\text{O}_4})} = \frac{(1.1)^2}{0.28} = 4.32$$

Now, volume is doubled, so pressure will become half and reaction proceeds in forward direction



$$\Rightarrow K_p = \frac{(0.55 + 2P)^2}{(0.14 - P)}$$

$$\Rightarrow 4.32 = \frac{(0.55)^2 + 2(0.55)(2P)}{0.14 - P}$$

$$= \frac{0.3025 + 2.20P}{0.14 - P}$$

(Ignoring P^2 terms)

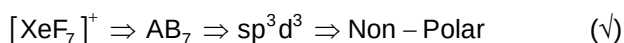
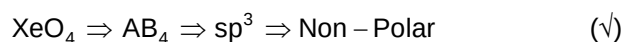
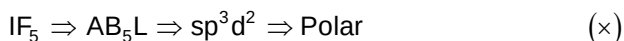
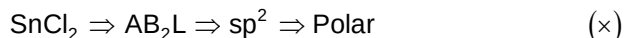
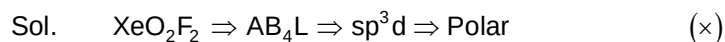
$$\Rightarrow P = 0.046 \text{ atm}$$

$$\therefore P_{\text{NO}_2} = 0.55 + 2 \times 0.046 \cong 0.642$$

$$= 64.2 \times 10^{-2} \text{ atm}$$

$$\Rightarrow x = 64.2 \sim 64$$

56. 2



57. 4

Sol. 1. Correct as EN of side atom increases, EN of carbon increases.

 2. Correct, $\text{Mg}^{++} \Rightarrow 3^{\text{rd}}$ period

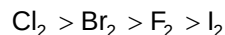
 $\text{K}^+ / \text{S}^{--} \Rightarrow 4^{\text{th}}$ period and isoelectronic

 $\text{Se}^{--} \Rightarrow 5^{\text{th}}$ period

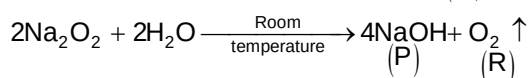
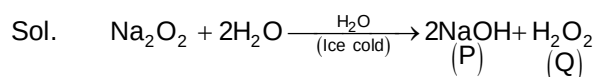
3. Correct as moving down the group cationic size increases so lattice energy decreases

 4. Correct as Hydration energy $\propto \frac{1}{\text{Size of cation}}$

5. Incorrect as the bond energy order among halogens is



58. 33



$$\text{Hence, } \frac{M_1 + M_2}{2} = \frac{34 + 32}{2} = 33$$

59. 80

 Sol. At half neutralization $|\text{S}| = |\text{A}|$

$$\text{pH} = \text{pK}_a + \log \frac{|\text{S}|}{|\text{A}|}$$

$$\Rightarrow \text{pH} = \text{pK}_a \Rightarrow \text{pK}_a = 5.0$$

Now, at equivalence point,

$$\text{PH} = \frac{1}{2}(\text{pK}_w + \text{pK}_a + \log C)$$

$$9 = \frac{1}{2}(14 + 5 + \log C)$$

$$\Rightarrow C = 0.1 \text{ M}$$

Let, V mL of NaOH is used

$$\text{Meq. (NaOH)} \Rightarrow \frac{0.25 \times V}{60 + V} = 0.1 = \text{Meq. (salt)}$$

$$\therefore \text{Volume (V)} = 40 \text{ mL}$$

$$\text{Meq. of acid} = \text{millimoles of acid} = 0.25 \times 40 = 10$$

$$\text{Mass of acid (HX)} = 10 \times 10^{-3} \times 82 = 0.829$$

$$\text{Mass\%} = \frac{0.82}{1.025} \times 100 = 80\%$$

60. 1

Sol. $[\text{H}^+]_{\text{total}} = 25 \times 0.1 + 50 \times 0.2 = 12.5 \text{ milimoles}$

$$[\text{OH}^-]_{\text{total}} = 25 \times 0.1 = 2.5 \text{ milimoles}$$

$$\therefore [\text{H}^+]_{\text{final}} = \frac{12.5 - 2.5}{25 + 50 + 25}$$

$$[\text{H}^+]_{\text{final}} = \frac{10}{100} \text{ M} = \frac{1}{10} \text{ M}$$

$$\therefore \text{pH} = -\log [\text{H}^+]_{\text{final}}$$

$$\text{pH} = -\log \left(\frac{1}{10} \right) = 1.00$$

Mathematics

PART – C

SECTION – A

61. C

Sol. $f'(x) + f(x) = 12x^2 + 2$

62. B

Sol. Use expansion and collect the coefficient of x^2

$$\Rightarrow P = 330$$

63. A

Sol. For A $x - \frac{x^3}{6} < \sin x < x$ and for B use limit as a sum

$$\Rightarrow A = \frac{1}{2}, B = \frac{2}{\pi}$$

64. D

Sol. $f'(x) = \frac{-1}{(\sin^2 x - (x-a)^2)^2} (\sin 2x - 2(x-a)) < 0 \quad \forall x \in (0, 1) \text{ and } a \in (2, 3)$

65. C

Sol. $A - B = \int_0^1 f(x) dx + \int_{f(0)}^{f(1)} f^{-1}(x) dx = 0$, and also $A = B$.

66. D

Sol. Domain is $\mathbb{R}$.

67. D

Sol. $A = \int_0^{\pi/4} (\tan x - \sin x) dx = \left(\frac{1}{2} \ln 2 + \frac{1}{\sqrt{2}} - 1 \right)$

68. A

Sol. $I_1 = \frac{\pi^2}{2} + 2, I_2 = \frac{\pi^2}{2} - 2, I_3 < 1$

69. D

Sol. $f'(x) = (\ln x)^2, g'(x) = (\ln x)$, so area $= \int_1^e (\ln x - (\ln x)^2) dx = 3 - e$

70. D

Sol. $L_1 = L_2 = \frac{\pi}{2}$

71. B

Sol. $T_n = \frac{((n+1)x+3) - (nx+3)}{((n+1)x+3)(nx+3)} = \frac{1}{nx+3} - \frac{1}{(n+1)x+3}$

$$S_{12} = V_1 - V_2 + V_2 - V_3 + \dots V_{12} - V_{13} = V_1 - V_{13}$$

$$= \frac{1}{x+3} - \frac{1}{13x+3}$$

$$\lim_{x \rightarrow 1} \frac{1}{4} - \frac{1}{16} = \frac{3}{16}$$

72. A

Sol. $\tan \theta = \frac{5}{12} = \frac{r}{h} \Rightarrow r = \frac{5h}{12}$

$$\frac{dv}{dt} = 0.1 \text{ m}^3/\text{min}$$

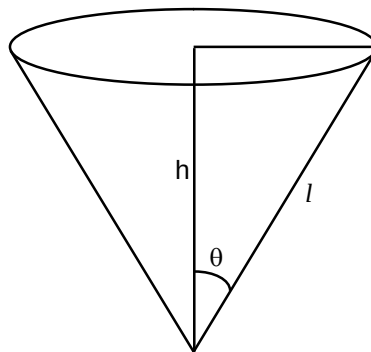
$$v = \frac{1}{3} \pi r^2 h = \frac{\pi 25 h^3}{432}$$

$$\Rightarrow \frac{dv}{dt} = \frac{25\pi h^2}{144} \frac{dh}{dt} \Rightarrow 0.1 = \frac{25}{144} \times \pi \times 3 \times 3 \times \frac{dh}{dt}$$

$$\Rightarrow \frac{16}{250} \pi = \frac{dh}{dt}$$

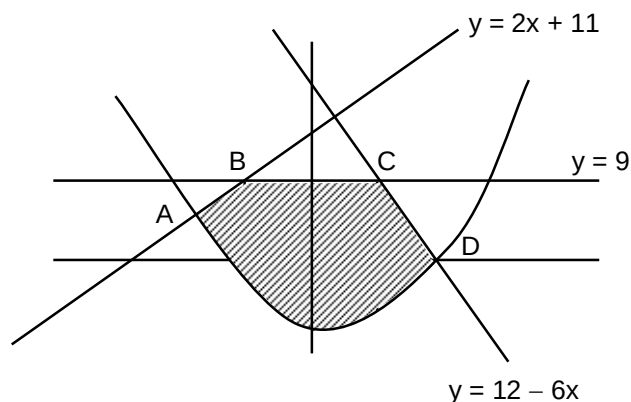
$$\text{Now } s = \pi r l = \pi r \sqrt{h^2 + r^2} = \frac{65\pi h^2}{144}$$

$$\frac{ds}{dt} = \frac{65\pi}{144} \times 2h \times \frac{dh}{dt} = \frac{65\pi}{144} \times 2 \times 3 \times \frac{16}{250\pi} = \frac{13}{75} \text{ m}^2/\text{min}$$



73. A

Sol. $x^2 - 4 = 2x + 11$
 $x = -3, 5 \therefore A(-3, 5)$
 $2x + 11 = 9$
 $x = 1 \therefore B(-1, 9)$
 $C\left(\frac{1}{2}, 9\right)$ and $D(2, 0)$



$$\text{Required area} = \int_{-3}^{-1} (2x+11) - (x^2-4) dx + \int_{-1}^{1/2} 9 - (x^2-4) dx + \int_{1/2}^2 12-6x - (x^2-4) dx$$

$$= \int_{-3}^{-1} (2x - x^2 + 15) dx + \int_{-1}^{1/2} 13 - x^2 dx + \int_{1/2}^2 (16 - 6x - x^2) dx$$

$$= x^2 - \frac{x^3}{3} + 15x \Big|_{-3}^{-1} + 13x - \frac{x^3}{3} \Big|_{-1}^{1/2} + 16x - 3x^2 - \frac{x^3}{3} \Big|_{1/2}^2$$

$$= \frac{40}{3} + \frac{153}{8} + \frac{81}{8} = \frac{511}{12}$$

74. B

Sol. $f'(t) = \frac{e^{t^2} t^5}{(t^4 + 2t^2 + 2)^2} \Rightarrow f'(1) = \frac{e}{25}$

$$\begin{aligned}
 \text{Let } x^2 = v &\Rightarrow \frac{dv}{dx} = 2x \Rightarrow \frac{1}{2} \int \frac{e^v v^2}{(v^2 + 2v + 2)^2} dv \\
 &\Rightarrow \frac{1}{2} \int e^v \left(\frac{1}{v^2 + 2v + 2} - \frac{2v + 2}{(v^2 + 2v + 2)^2} \right) = \frac{1}{2} \frac{e^v}{v^2 + 2v + 2} \\
 &= \frac{1}{2} \frac{e^{x^2}}{x^4 + 2x^2 + 2} \\
 \therefore f(t) &= \frac{1}{2} \frac{e^{x^2}}{x^4 + 2x^2 + 2} \Big|_0^t = \left(\frac{e^{t^2}}{t^4 + 2t^2 + 2} - \frac{1}{2} \right) \frac{1}{2} \\
 \therefore f(1) &= \frac{e}{10} - \frac{1}{4} \\
 \therefore f(1) + f'(1) &= \frac{e}{10} + \frac{e}{25} - \frac{1}{4} = \frac{7e}{50} - \frac{1}{4}
 \end{aligned}$$

75. B

Sol. $\lim_{x \rightarrow \infty} \frac{(k-2)x^3 + (3+n)x^2 + (m-3)x + (2-p)}{\sqrt{x^4 + kx^3 + 3x^2 + mx + 2} + \sqrt{x^4 + 2x^3 - nx^2 + 3x - p}}$

$k-2$ has to be 0 else limit wont be finite

$$\Rightarrow k = 2$$

Now dividing Num. & Den by x^2

$$\begin{aligned}
 &\lim_{x \rightarrow \infty} \frac{(3+n) + \frac{m-3}{x} + \frac{(2-p)}{x^2}}{\sqrt{1 + \frac{a}{x} + \frac{3}{x^2} + \frac{b}{x^3} + \frac{2}{x^4}} + \sqrt{1 + \frac{2}{x} - \frac{c}{x^2} + \frac{3}{x^3} - \frac{d}{x^4}}} \\
 &= \frac{3+n}{2} \Rightarrow \frac{3+n}{2} = 4 \Rightarrow n = 5.
 \end{aligned}$$

76. A

Sol. Let $x^2 = t \Rightarrow \frac{dt}{dx} = 2x \Rightarrow I = \frac{1}{2} \int_0^{2\pi} \frac{t \sin^{2022} t \, dt}{\cos^{2022} t + \sin^{2022} t}$

$$\begin{aligned}
 \Rightarrow I &= \frac{1}{2} \int_0^{2\pi} \frac{(2\pi - t) \sin^{2022} t}{\cos^{2022} t + \sin^{2022} t} \Rightarrow 2I = \int_0^{2\pi} \frac{\pi \sin^{2022} t}{\cos^{2022} t + \sin^{2022} t} \\
 \Rightarrow 2I &= 4 \int_0^{\frac{\pi}{2}} \frac{\pi \sin^{2022} t}{\cos^{2022} t + \sin^{2022} t} \\
 \Rightarrow 2I &= 4 \int_0^{\frac{\pi}{2}} \frac{\pi \cos^{2022} t}{\cos^{2022} t + \sin^{2022} t} \\
 \Rightarrow 4I &= 4 \int_0^{\frac{\pi}{2}} \pi \, dt \Rightarrow I = \frac{\pi^2}{2}.
 \end{aligned}$$

77. A

Sol. $\frac{dy}{dx} = \frac{(x-2)+(y-2)}{(x-2)-(y-2)}$

put $x-2 = h$, $y-2 = k$

$$\frac{dk}{dh} = \frac{h+k}{h-k} \quad \text{put } k = vh$$

$$\Rightarrow v + h \frac{dv}{dh} = \frac{1+v}{1-v}$$

$$\Rightarrow \int \frac{1-v}{1+v^2} dv = \int \frac{dh}{h} \Rightarrow \tan^{-1} v = \frac{1}{2} \ln(1+v^2) + \ln h + c$$

$$\Rightarrow \tan^{-1} \left(\frac{y-2}{x-2} \right) = \frac{1}{2} \ln \left(1 + \frac{(y-2)^2}{(x-2)^2} \right) + \ln(x-2) + c$$

$\therefore$ it passes through (3, 2)

$$\Rightarrow \tan^{-1} 0 = \frac{1}{2} \ln 1 + \ln 1 + c \Rightarrow c = 0$$

$\therefore$ it also passes through (p+2, 3)

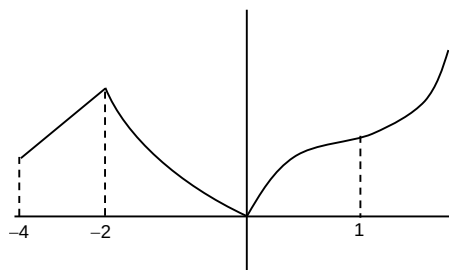
$$\tan^{-1} \left(\frac{1}{p} \right) = \frac{1}{2} \ln \left(1 + \frac{1}{p^2} \right) + \ln p$$

$$\Rightarrow 2 \tan^{-1} \left(\frac{1}{p} \right) = \ln(1+p^2)$$

78. B

Sol. $A = \int_{-4}^{-2} (x+6) dx + \int_{-2}^0 x^2 dx + \int_0^1 \sqrt{x} dx + \int_1^2 x^2 dx$

$$= 6 + \frac{8}{3} + \frac{2}{3} + \frac{7}{3} = \frac{35}{3}$$



79. C

Sol. $f(e^4) = \int_1^{e^4} \frac{\log_{10} x}{1+x} dx \quad \dots(1)$

Let $x = \frac{1}{v} \Rightarrow \frac{dx}{dv} = \frac{-1}{v^2}$

$$\Rightarrow f(t) = \int_1^{1/t} \frac{\log_{10}(1/v)}{1 + \frac{1}{v}} - \frac{1}{v^2} dv = \int_1^{1/t} \frac{\log_{10} v}{v(v+1)} dv$$

$$\Rightarrow f(t) = \int_1^{1/t} \frac{\log_{10} v}{v(v+1)} dv \Rightarrow f(e^{-4}) = \int_1^{e^4} \frac{\log_{10} v}{v(v+1)} dv = \int_1^{e^4} \frac{\log_{10} x}{x(x+1)} dx \quad \dots(2)$$

Add (1) and (2)

$$f(e^{-4}) + f(e^4) = \frac{1}{\ln 10} \int_1^{e^4} \frac{\ln x}{(1+x)} \left(1 + \frac{1}{x} \right) dx$$

$$= \frac{1}{\ln 10} \int_1^{e^4} \frac{\ln x}{x} = \frac{1}{\ln 10} \left[\frac{(\ln x)^2}{2} \right]_1^{e^4}$$

$$= \frac{1}{2 \ln 10} 16 = \frac{8}{\ln 10}$$

80. C

Sol. Let $g(x) = x^3 - 2x^2 + 3x - 9$
 $g'(x) = 3x^2 - 4x + 3 > 0 \quad \forall x \in \mathbb{R}$
 $\Rightarrow g(x)$ is always increasing
 $\therefore g(2) = 8 - 8 + 6 - 9 = -3$ is maximum value for $x \leq 2$
 If $-3x + \log^3(k^2 - 5) \leq -3$ for $x > 2$
 then we have maximum at $x = 2$
 $\Rightarrow -6 + \log^3(k^2 - 5) \leq -3 \Rightarrow \log^3(k^2 - 5) \leq 3$
 $\Rightarrow k^2 - 5 \leq 27 \Rightarrow (k - 4\sqrt{2})(k + 4\sqrt{2}) \leq 0$
 $k \in [-4\sqrt{2}, 4\sqrt{2}]$ and $k^2 - 5 > 0$
 $\Rightarrow k \in (-\infty, -\sqrt{5}) \cup (\sqrt{5}, \infty)$
 $\therefore k \in [-4\sqrt{2}, -\sqrt{5}) \cup (\sqrt{5}, 4\sqrt{2}]$

SECTION - B

81. 1

Sol. $\left[\frac{x^2 + x + 2}{x^2 + x + 1} \right] = 1$

82. 1

Sol. $A = \frac{\pi^3}{8} - \frac{\pi^2}{4}$

83. 4

Sol. $D_f = [-2, 2]$ as $f(x) = \sqrt{3}(\sqrt{4 - x^2})$
 $R_f = \in [0, 2\sqrt{3}]$

84. 2

Sol. $f(x) = \cos(x^2 - 3x) |x - 2| \sin|x - 2| + (x - 1) |x| |x - 1| |x - 7|$
 $\therefore f(x)$ non differentiable at $x = 0$ and 7 only.

85. 12

Sol. $a = 1, b = 6, c = 5$

86. 96

Sol. Put $x = 0$ we get $y = 1$
 $\ln(x + y) = 6xy$
 Differentiate w.r.t. x
 $\frac{1}{x + y} \left(1 + \frac{dy}{dx} \right) = 6 \left(\frac{xdy}{dx} + y \right)$

$$\text{at } x = 0, y = 1 \Rightarrow \frac{dy}{dx} = 5$$

$$\Rightarrow 1 + \frac{dy}{dx} = 6(x + y) \left(x \frac{dy}{dx} + y \right)$$

Differentiate w.r.t. x both side

$$\begin{aligned} \Rightarrow \frac{d^2y}{dx^2} &= 6 \left((x + y) \left(\frac{xd^2y}{dx^2} + \frac{dy}{dx} + \frac{dy}{dx} \right) + \left(\frac{xdy}{dx} + y \right) \left(1 + \frac{dy}{dx} \right) \right) \\ &= 6(1(0 + 5 + 5) + 1(6)) = 96 \end{aligned}$$

87. 9

Sol. $\therefore \lim_{x \rightarrow 0} \left(3 - 2\cos x \sqrt{\cos 2x} \right)^{\left(\frac{x+3}{x^2} \right)}$ is of the form 1^∞

$\therefore$ it is equal to $e^{\lim_{x \rightarrow 0} \left(\frac{2 - 2\cos x \sqrt{\cos 2x}}{x^2} \right) (x+3)}$

$$\begin{aligned} \text{Now, } \lim_{x \rightarrow 0} \frac{2 - 2\cos x \sqrt{\cos 2x}}{x^2} &= \lim_{x \rightarrow 0} \frac{\frac{-2\cos x}{2\sqrt{\cos 2x}} \times -\sin 2x \times 2 - \sqrt{\cos 2x} (-\sin x)}{2x} \\ &= \lim_{x \rightarrow 0} \frac{\frac{2\cos x \sin 2x}{\sqrt{\cos 2x}} + 2\sin x \sqrt{\cos 2x}}{2x} \\ &= 2 + 1 = 3 \end{aligned}$$

$$\begin{aligned} \therefore e^{\lim_{x \rightarrow 0} \left(\frac{2 - 2\cos x \sqrt{\cos 2x}}{x^2} \right) (x+3)} &= e^{3 \times 3} = e^9 \therefore a = 9 \end{aligned}$$

88. 14

Sol. $\frac{dy}{dx} = 3x^2 - 4x + 4 \Rightarrow \left(\frac{dy}{dx} \right)_{(a,b)} = 3a^2 - 4a + 4$

$$\frac{dy}{dx} = 6x + 2 \Rightarrow \left(\frac{dy}{dx} \right)_{(1,4)} = 8$$

$$\Rightarrow 3a^2 - 4a + 4 = 8 \Rightarrow a = 2, -\frac{2}{3}$$

$$\therefore b = 8 - 8 + 4 = 4$$

89. 144

Sol. $\int \frac{12x^{15} + 5x^{18}}{x^{24}(-7 + 3x^{-8} + 2x^{-5})^3} dx = \int \frac{12x^{-9} + 5x^{-6}}{(-7 + 3x^{-8} + 2x^{-5})^3}$

$$\text{Let } -7 + 3x^{-8} + 2x^{-5} = t \Rightarrow \frac{dt}{dx} = -24x^{-9} - 10x^{-6}$$

$$= -\frac{1}{2} \int \frac{1}{t^3} dt = -\frac{1}{2} \frac{t^{-2}}{-2} + c = \frac{1}{4(-7 + 3x^{-8} + 2x^{-5})^2} + c$$

$$\therefore f(1) = \frac{1}{16} \Rightarrow \frac{1}{16} = \frac{1}{4(-7 + 3 + 2)^2} + c \Rightarrow c = 0$$

$$\therefore f(x) = \frac{1}{4(-7 + 3x^{-8} + 2x^{-5})^2} \quad \therefore f(-1) = \frac{1}{4(-6)^2} = \frac{1}{144}$$

90. 23

Sol. $I_n = \int_1^e x^{23} \left(\log x \right)^n dx$

$$\Rightarrow I_n = (\ln x)^n \frac{x^{24}}{24} \Big|_1^e - \int_1^e n \frac{(\ln x)^{n-1}}{x} \frac{x^{24}}{24} dx$$

$$\Rightarrow I_n = \frac{e^{24}}{24} - \frac{n I_{n-1}}{24} \Rightarrow 24 I_n = e^{24} - n I_{n-1}$$

Put $n = 15$ $24 I_{15} = e^{24} - 15 I_{14}$... (1)

Put $n = 14$ $24 I_{14} = e^{24} - 14 I_{13}$... (2)

$$24 I_{15} - 24 I_{14} = 14 I_{13} - 15 I_{14}$$

$$24 I_{15} = 9 I_{14} + 14 I_{13}$$